Sprint #2 (2/13 – 2/20)

Scrum Master: Palwasha Newell

Project:

ETSU Marketplace is a university-only online marketplace that allows verified students to buy, sell, and exchange items and housing opportunities securely. It is a marketplace for students and only students to use. Built in ASP.NET MVC and C#.

Activities

Sprint 2 focused on implementing core marketplace functionality and transitioning from mock/test data to dynamic database-driven content. During this sprint, we continued holding Scrum meetings during class sessions and used Discord for communication outside of class. We refined backlog items at the beginning of the sprint, assigned story points internally (though not formally tracked), and moved selected items into the Sprint Backlog.

Unlike Sprint 1, which focused heavily on UI structure and foundational setup, Sprint 2 centered on implementing CRUD functionality, API controllers, database integration, search capabilities, homepage listing population, and messaging functionality. We worked extensively with Entity Framework Core, repository patterns, API controllers, and search filtering logic. We also began integrating backend functionality with frontend components to ensure listings displayed dynamically rather than using placeholder test data. A Sprint Review and Sprint Retrospective were held at the end of the sprint to evaluate progress and identify workflow improvements.

Challenges

One challenge during Sprint 2 was transitioning from static test data to a fully database-driven implementation. Configuring DbContext correctly and ensuring repositories properly interacted with the database required additional debugging time. Another challenge was coordinating frontend and backend integration. While APIs were functional, ensuring correct data binding to views and partial views required adjustments. Messaging implementation introduced additional complexity due to authentication integration and secure user-to-user communication requirements. Despite these technical challenges, the team was able to collaborate effectively and complete all Sprint 2 goals.

Achievements

By the end of Sprint 2, our marketplace evolved from a UI prototype into a functioning application capable of:

- Creating listings with persistent database storage
- Viewing active listings dynamically from the database
- Displaying listings on the homepage as cards
- Searching listings
- Contacting sellers through the platform

We successfully implemented API controllers, repository patterns, and expanded our database structure. The application now supports core marketplace functionality and is ready for further enhancements in filtering, editing, deleting, and advanced messaging features.

Member Contributions & Owned Tasks

Jacob Sullins

User Story Completed:
• *As a student, I want to create a listing with a title, description, price, and images so that I can sell an item to other students.*

Work Completed:

- Created **LeaseListingAPIController**
- Created **ItemListingAPIController**
- Implemented Listing Repositories (Repository Pattern)
- Configured and updated **DbContext** to handle Listings
- Connected listing creation to persistent database storage

Estimated Time Spent:

Task	Estimated Time
Designing API Controllers	4 hours
Implementing Repository Pattern	5 hours
Configuring DbContext & migrations	3 hours
Integrating listing creation with UI	4 hours
Debugging & Testing	3 hours
Total Estimated Time	**19hours**

This task required substantial backend architecture work and database integration, making it the most time-intensive contribution of the sprint.

Jabari Mitchell

User Story Completed:
• *As a student, I want to see listings as cards on the home page so that I can browse items on the marketplace.*

Work Completed:

- Modified HomeController to retrieve active listings from the database
- Connected listing repositories to homepage logic
- Integrated listing card partials with live data
- Ensured proper Bootstrap styling and layout consistency

Estimated Time Spent:

Task	Estimated Time
Controller updates for dynamic data	3 hours
Repository integration	3 hours
Frontend binding & Razor adjustments	3 hours
Testing & layout debugging	2 hours
Total Estimated Time	**11 hours**

This contribution transitioned the homepage from static mock data to dynamic marketplace content.

Nathanael Wolfe

User Stories Completed:
• *As a student, I want to contact a seller through the platform so that I can ask questions about their lease takeover request.*
• *As a student, I want to contact a seller through the platform so that I can ask questions about an item.*

Work Completed:

- Implemented messaging/contact functionality
- Integrated messaging logic with ASP.NET Identity
- Ensured messages are associated with correct listings and users
- Implemented backend handling for sending and receiving messages
- Secured communication endpoints

Estimated Time Spent:

Task	Estimated Time
Designing messaging flow	3 hours
Backend implementation	3 hours
Identity integration	3 hours
Testing communication logic	3 hours
Debugging & validation	2 hours
Total Estimated Time	**13 hours**

Messaging required careful authentication integration and backend logic to ensure secure communication between users.

Palwasha Newell

User Stories Completed:
• *As a student, I want to view all active listings so that I can browse items available for sale.*
• *As a student, I want to search for listings so that I can find more specific items.*

Work Completed:

- Implemented logic to retrieve all active listings from the database
- Connected ListingsController to repository layer
- Built search functionality using query parameters
- Implemented search filtering in controller logic
- Updated views to dynamically render filtered results

Estimated Time Spent:

Task	Estimated Time
Implementing active listings retrieval	3 hours
Repository integration	2 hours
Search logic implementation	2 hours
View updates & Razor binding	2 hours
Testing search functionality	2 hours
Total Estimated Time	**11 hours**

This work significantly improved user experience by allowing browsing and targeted searching.

Team Velocity

For Sprint 2, although formal story points were still not strictly assigned, we completed:

- 6 Major User Stories
- Multiple backend infrastructure tasks
- Full database integration for listings

Estimated total team development time:
54 combined hours
Average per team member:
13-14 hours
Compared to Sprint 1 (which was lighter and more foundational), Sprint 2 showed a clear increase in workload and productivity. We established a stronger backend foundation and moved closer to a fully operational marketplace system.